

The Multimodal Exhibition: Tangible Interactions for People With Visual Impairments

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Abstract

People-with-visual-impairments (PVI) face persistent barriers in interacting with the built environment because design continues to prioritize use of sight. This paper presents a human-centered, multimodal exhibition designed to evaluate six tangible prototypes aimed at enhancing awareness of surroundings, visualization-skills, and tactile-vocabulary for PVI and sighted participants. Prototypes ranged from tactile-sketchboards and audio-enabled-devices to cultural-exploration-tools tested in a guided dark-room experience called the Multimodal Box. Qualitative-interviews and behavioral-observations were used to explore sensory perception, prototype usability and empathy-driven insights. One hundred participants including 35 PVI, engaged with the experience. Post-exhibition experience interviews revealed significant shifts in participants' spatial-awareness and understanding of non-visual-navigation. Notably, over 90% of PVI found at least two prototypes applicable in daily life—Tactosketchboard and Magic Platform. By evaluating tangible-interventions in a near-realistic setting, this research contributes to the discourse on inclusive interaction design—encouraging co-designed systems that empower PVI to independently engage with their world through multi-sensory modalities.

Keywords

people with visual impairments, tangible interactions, multimodal interactions, spatial memory, multisensory design, touch, sound, system usability scale, assistive technologies, inclusive interactions

Introduction

India has 35 million people-with-visual-impairments (PVI), including 4.95 million people who are blind (NPCBVI, 2019). Despite programs like the Accessible India Campaign (2015) and enactment of Rights of Persons with Disabilities (RPwD) Act (2016), PVI have had to struggle with inaccessible public infrastructure, lack of inclusive education and exclusion from the job market. From a legal standpoint, the RPwD Act (2016) mandates universal accessibility in transportation, education, public spaces, and digital platforms. However, implementation of the RPwD Act has been inconsistent. Sidewalks remain cluttered, public signages lack tactile or auditory cues, and digital interfaces prioritize visual layout over accessibility. As Sehrawat and Suvitha (2020) elucidate, “infrastructure and policy, while progressive in language, often fail to translate into independent navigation or participation for PVI.” Hersh and Johnson (2008) argue for a shift from assistive to inclusive technology, suggesting that the focus should not just be on functional independence but also on social participation and identity. Recent works in

inclusive interaction design explore multisensory modalities—where touch, sound, and spatial memory come together to create the inclusive user experience (Ladner, 2015; Giudice et al., 2019).

Multimodal interaction has proven effective in supporting learning, exploration, and communication for users with sensory impairments (Samaniego López et al., 2025). Moreover, the design process typically excludes people with disabilities, leading to what Pullin (2009) describes as a “gap between usability and desirability.” Therefore, this project addresses these gaps by integrating tactile, auditory and olfactory cues in a shared exhibition space designed for PVI and sighted

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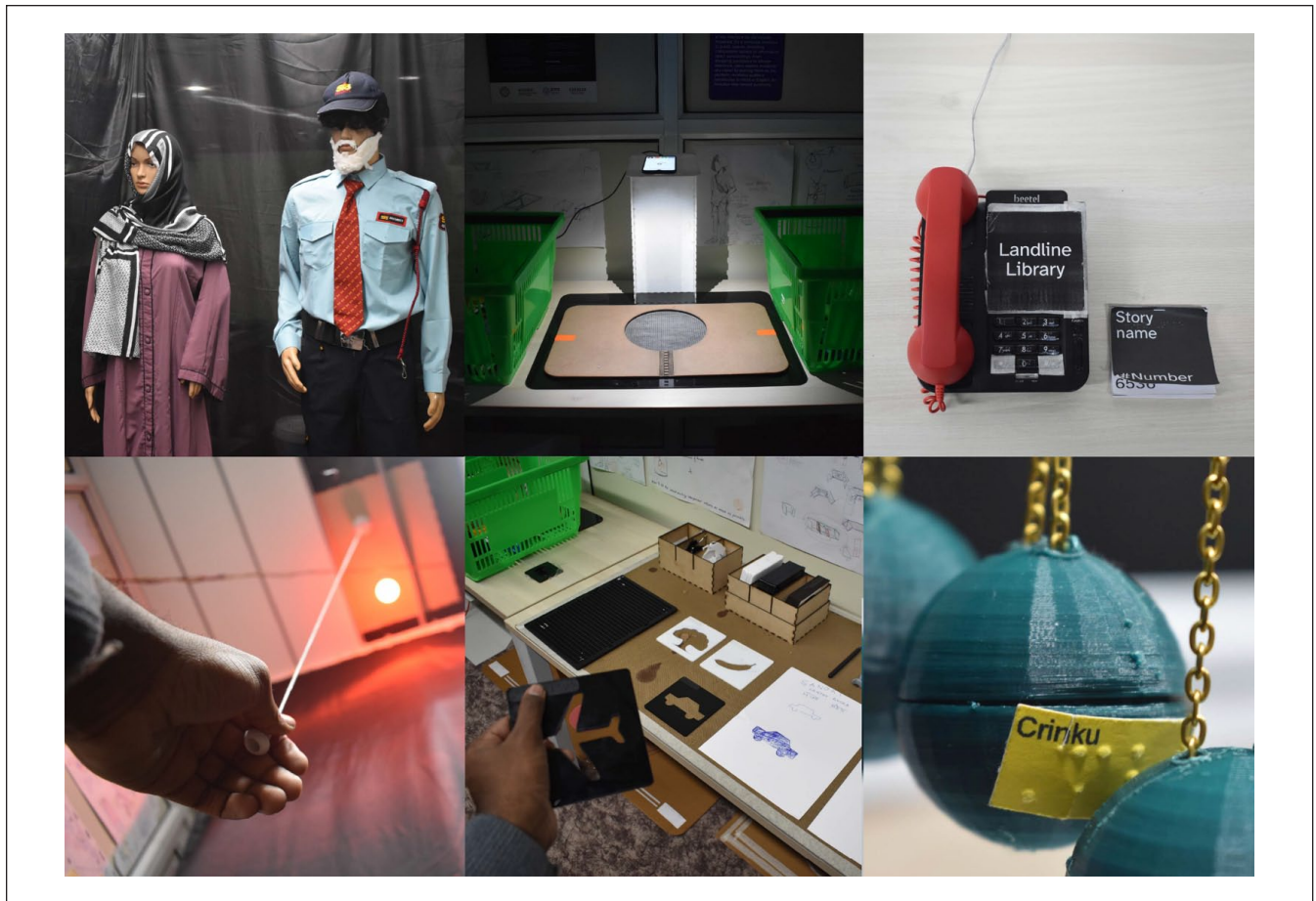


Figure 1. Clockwise from top left: 1. Touching cultures; 2. Magic platform; 3. Landline-phone library; 4. Sound souvenirs; 5. Tactosketchboard; and 6. Pull vocabulary.

participants. The goal was not merely to assist users in a world that is primarily designed for visual inputs but to reimagine interactions that prioritize non-visual intelligence as an asset. Importantly, this project reframes assistive technology as interactional design, situating accessibility not as an add-on, but as a foundational premise.

Six Tangible Interactions

This article presents design, implementation, and evaluation of six tangible interactions, each addressing a distinct challenge faced by PVI. The project is rooted in a simple yet powerful idea—instead of creating assistive technologies that retrofit a visual world, can we reimagine interactions that are naturally multimodal, intuitive and shared across users varying in their abilities? Six prototypes—Touching Cultures, Magic Platform, Landline Phone-Library, Sound Souvenirs, Tactosketchboard, and Pull Vocabulary were showcased in a darkroom simulating full vision loss (Figure 1). The immersive format enabled PVI and sighted users to experience interactions on equal terms, revealing key insights into perception, agency, and inclusion.

Touching Cultures

Sighted people learn a lot about grooming, dressing, cultural patterns and fashion trends by observing others visually. Such experiences are very limited for PVI. We used portable mannequins (male and female), arranged hair wigs of different styles and added various types of beards like full, French, mutton chops, and soul patch using cotton strips. The mannequins enabled PVI to experience and understand tangible clothing characteristics on fully dressed and groomed mannequins without awkwardness of touching the opposite gender. The mannequins portrayed various kinds of professions, religions, celebrities, etc., giving a first-time experience of understanding what majority have imagined through verbal descriptions till this very point in life.

Magic-Platform

A portable and hands-free accessible kiosk for independent exploration and easy identification of any tagged product. Comprises a platform with textured markers and an inverted L-shaped stand with a mobile phone mounted on top with the

camera facing downwards. Products are tagged with dual-audio QR codes—each tag has a smooth side (triggering English audio) and a rough side (triggering Hindi audio). Users place the tag on the textured platform, and the camera scans it to play the respective audio description.

Landline Phone-Library

This uses a dummy landline phone with a Raspberry-Pi running inside it. Users can press four digits on a number-pad equipped with braille-embossed stickers to listen to a short story through the handset. The list of stories with 4-digit codes is printed like a phonebook with information in text and braille. Users can refer to this directory, punch the respective 4-digit code and listen to the audio-file of the story. Such phones can be placed in various corners of a school for PVI, creating accessible spaces where children can listen and enjoy stories on their own.

Sound Souvenirs

A sighted girl recognizes her mother from a distance just by looking. A girl with visual impairment can do the same by listening to the sound of bells from her mother's anklet or bracelet. This scenario inspired creation of low-cost pendant-sized sound souvenirs. We made 3D-printed pendants of different shapes and filled them with edible grains, small pebbles and seashells to create unique sound recipes from soft crinkles to calming bells. Users can create their personalized sounds and can wear the pendants like a necklace or as a wrist bracelet. These sound souvenirs act as a form of an identity in the PVI community like we have smell or fashion identities. It enables them to identify a person with auditory information, even in non-verbal scenarios

Tactosketchboard

An A4-sized drawing board that has a rectangular mesh-like texture on the drawing surface. PVI can draw with a ball-point pen/stylus by placing paper over the mesh surface. Additionally, we developed stencils for 20+ shapes which can be placed on paper, then the shape can be traced, and the area inside scribbled to get an embossed impression of the 2D objects on the reverse side. This tactile sketch board gave PVI the opportunity to draw, play, and express themselves, many for the first time in their lives.

Pull-Vocabulary

An audio-interface triggered by a roof mounted pull-cord switch that triggers an audio-output upon pulling the hanging cord, like the animal's name, followed by its sound (e.g., Lion "Roaring sound"). Audio files are stored on a Raspberry Pi and the output sent through a speaker. It can be installed in classrooms and any educational content can be integrated.

The audio-interface enables a fun, interactive activity-based approach to learning for students with visual impairments.

Inclusive by Design

Our broader aim was to catalyze new conversations about how tangible interfaces can evolve beyond accessibility toward co-designed systems where PVI are not just recipients but key contributors to the innovation. Through observational data, semi-structured interviews, and multimodal data synthesis techniques, the study offered a lens into how tactile, auditory and olfactory can together build richer and more intuitive experiential environments. The study investigated how such experiences can also drive awareness among sighted participants, thus mainstreaming universal design of tangible interactions. Importantly, including blindfolded sighted participants was a deliberate methodological choice to simulate real-world scenarios of sudden or progressive vision loss—such as late-onset blindness or age-related conditions. Representing those who may lose vision later in life, their disorientation, adaptive responses, and emotional reactions offered critical insights into how inclusive interfaces can better support this growing demographic. According to global projections, over half a billion people are expected to experience moderate to severe vision loss by 2050, much of it in later life stages (Flaxman et al., 2021).

Methodology

The study followed a two-phase research process. First phase focused on fieldwork, data synthesis and design development. Second phase involved evaluation of the tactile prototypes by PVI and blindfolded sighted users in a dark-room exhibition using qualitative and semi-quantitative methods.

Phase 1: Primary Research and Multimodal Synthesis

The initial research involved extensive engagement with PVI in schools and assistive technology centers in New Delhi and Hyderabad, India. To analyze this qualitative data, we developed a multimodal mind-mapping technique—a three-layer data synthesis process combining raw observation, interpretive insight, and ideation. This method, described in prior work (Koruprolu & Kar, 2024), enabled us to synthesize research data into actionable design concepts.

Phase 2: Exhibition-Based Evaluation

To evaluate usability, intuitive appeal and potential real-world impact of the prototypes, we curated a guided dark-room experience called The Multimodal Box. This immersive setup enabled PVI and the sighted to experience six prototypes non-visually with touch, sound, and smell (Figure 2).



Figure 2. A view of the multimodal box with the six prototypes.

Participants

A total of 100 participants attended the exhibition—25 with complete vision loss, 10 with partial vision loss and 65 sighted participants who experienced the exhibition wearing blindfolds.

Structure of the Evaluation

Pre-Experience Interviews (Only for the Sighted). Before entering the dark room, sighted participants were asked: (1) *Have you interacted with a PVI before?* (2) *How do you imagine they navigate everyday tasks?* (3) *What comes to your mind when you think of accessibility?* These questions helped gauge the baseline awareness and assumptions of the sighted participants.

Experience in the Dark Room (All Participants). Participants navigated the space either blindfolded (if sighted) or naturally (if visually impaired). The room was designed with tactile, auditory and olfactory orientation cues such as Mario soundtrack and distinct attars at each prototype. Participants explored each interaction one-by-one, guided by finger snaps and verbal instructions, without visual input.

Post-Experience Interviews (Blindfolded for the Sighted). Immediately after the experience, participants answered semi-structured questions which included the following: (1) *What*

are you feeling right now? (2) *Which two prototypes stood out for you and why?* (3) *Which of these would be helpful in your daily life?* (4) *How did your senses engage through the experience?* (5) *Would you use/recommend these solutions?* All responses were recorded for thematic analysis.

Final Reflection (Blindfold Removed for the Sighted). Participants were reintroduced to the same room—now without blindfolds and brightly illuminated. They were asked: (1) *Is the space how you imagined it?* (2) *Did any perceptions surprise you?* (3) *Now that you can see the prototypes, do you think your experience has changed?* (4) *What details did you miss without vision?* This allowed for a comparative reflection between imagined (non-visual) and actual (visual) spatial understanding. Interview transcripts were thematically coded using grounded analysis principles (Braun & Clarke, 2006), allowing for emergent patterns related to sensory engagement, independence, spatial adaptation, and emotional impact.

Quantitative Measures

Though primarily qualitative, the study also incorporated quantitative feedback described below:

- Prototype usability rating (1–5 scale for each).
- Relevance to daily life (5-point Likert scale from strongly disagree to strongly agree).

Table 1. Results of the Structured Prototype Evaluation.

Prototype	Avg. usability (1–5)	Daily life relevance (% agree)	Ranked in top 2 (%)
Magic platform	4.7	92%	61
Tactosketchboard	4.6	96%	72
Landline library	3.7	69%	38
Touching cultures	3.9	74%	32
Pull vocabulary	4	66%	29
Sound souvenirs	4.2	78%	45

- Top 2 prototype ranking (based on user perception).
- System Usability Scale (SUS)—10-item, 0 to 100 scale.

Findings

The Multimodal Box served as both an evaluative testbed and an experiential learning space. Findings were drawn from the observational notes, pre-and post-experience interviews, and structured feedback scores from all participants. Results are presented in two parts: (1) Thematic insights from qualitative data and (2) Prototype evaluation results from quantitative measures.

Thematic Insights from Qualitative Data

Spatial Disorientation and Sensory Adaptation. Sighted participants reported feeling disoriented without vision. When they removed their blindfolds, 9 out of 10 were surprised by how simple the room was—highlighting the distortions in their mental maps. In contrast, PVI navigated more easily, relying on tactile cues with ease. This affirmed the value of intuitive, hands-on design.

Reclaiming Expression Through Drawing. The Tactosketchboard received emotional feedback, especially from PVI participants who had never drawn before. A 25-year-old woman created a complete scenery using stencils and tactile sketching—her first drawing experience. This prototype was seen not only as a learning aid but also as a tool for identity and creative self-expression, validating the project's goal of going beyond utility toward empowerment.

Identity and Familiarity Through Sound. Sound Souvenirs was especially popular for its ability to encode emotional memory into a sound-based identifier. Sighted participants responded positively to the idea, while PVI users appreciated the choice to personalize. The prototype hinted at broader potential for wearable sensory augmentation.

Relevance of Multimodal Feedback. Participants consistently preferred prototypes that combined multiple modalities: touch and sound, or texture and form, over single-sensory designs. For example, the Magic Platform paired tactile

markers with audio labels to help participants identify objects. Multimodal feedback helped to overcome information gaps typically encountered in activities such as shopping, education, or browsing unfamiliar items.

Structured Prototype Evaluation Results

The Magic Platform and Tactosketchboard emerged as clear favorites, with both PVI and sighted users citing their ease of use and real-life applications (Table 1). While Sound Souvenirs scored modestly in all metrics, it showed a high emotional resonance, especially in open-ended responses. Touching Cultures and Pull Vocabulary, while less favored individually, were appreciated for their use in context of group learning activities. Tools that combined independence with emotional or expressive value ranked highest. The success of the prototype was not just based on usability, but also on how participants felt seen, capable, and engaged.

System Usability Scale (SUS) Evaluation

The SUS evaluation provided insights into the perceived usability and emotional appeal of each prototype (Bangor et al., 2008; Lewis, 2018; Table 2). Tactosketchboard received the highest average score (90), reflecting not only its intuitive design but also the emotional impact on users, who were drawing for the first time. Magic Platform also scored highly, driven by its multimodal feedback. Interestingly, Sound Souvenirs performed slightly lower in usability, but participant comments suggested it held deep personal value—hinting that emotional relevance may not always translate into raw usability scores. The low scores for Touching Cultures reflected a need for clearer instructions or a better onboarding process.

Discussion

Reframing Assistive Technology as Inclusive Interactions

A critical insight emerging from this project is the distinction between designing for PVI versus designing with and around their lived sensory realities. Traditionally, assistive technology aims to retrofit environments designed for the sighted. This

Table 2. Results of the System Usability Scale (SUS).

Prototype	Avg. SUS score	Key takeaways
Magic platform	87	Very high usability, especially appreciated for shopping, learning, and identifying objects. Strong multisensory alignment
Tactosketchboard	90	Highest emotional engagement. Enabled self-expression, intuitive use even for first-time drawers
Sound souvenirs	76	Emotionally powerful but slightly less intuitive. High personalization potential for PVI users
Landline library	68	Moderate usability. Recognized more for group utility. Interface could be improved for clarity
Touching cultures	70	Good for tactile exploration and cultural learning. Some users needed assistance initially
Pull vocabulary	73	Simple and clear. Most valuable in classroom or group-based vocabulary building

paradigm reinforces dependence and marginalization (Ladner, 2015). In contrast, our approach treats non-visual perception as a valid design starting point—not as a constraint. Prototypes like the Magic Platform and Tactosketchboard exemplify this reframing. The Magic Platform doesn’t just “label” objects—it reimagines how product discovery could work in schools, libraries and stores. Likewise, the Tactosketchboard reclaims drawing as not just a learning tool but a medium for emotional self-expression—something often denied to the PVI due to assumptions around visual literacy. These outcomes align with Pullin’s (2009) argument that design should move beyond usability and consider desirability and dignity.

Insights for Multisensory Design Principles

Findings highlight the best practices for multi-sensorial interaction design:

- *Redundancy across modalities:* Prototypes that combined tactile and auditory feedback were consistently rated higher. This supports previous research showing that multisensory redundancy improves usability and comprehension among PVI (Götzelmann, 2012).
- *Simplicity of feedback:* Participants appreciated designs where feedback was clear, predictable, and scaffolded (e.g., pressing a button, pulling a cord, tracing a stencil). Complex or ambiguous feedback reduced engagement.
- *Personalization:* Sound Souvenirs, though less functional, resonated deeply due to its potential for identity creation. This suggests that personalization through audio cues or modular designs should be prioritized in future designs.

Importantly, these lessons extend beyond disability. As Samaniego López and colleagues (2025) argue, multimodal design improves usability for everyone—from aging users to children learning new tasks—making these prototypes strong candidates for universal design adaptation. Findings from sighted participants echo Giudice et al.’s (2019) research on sensory substitution and highlight how non-visual navigation can be supported through environmental scaffolds.

Conclusion

Two prototypes—the Tactosketchboard and Magic Platform—emerged as strong candidates for real-world deployment, owing to their simplicity, affordability, and everyday relevance. Their success was evident not just in usability scores, but in how they redefined agency. One participant, who had never drawn before, did so with joy and pride; another imagined navigating a store and buying clothes independently for the first time. These moments highlight the deeper goal of inclusive design: empowerment, not just accommodation. The dark-room experience—The Multimodal Box—offered a rare chance to simulate and reflect on non-visual perception. It revealed how sighted participants—especially designers, educators, and policymakers—benefit from tangible firsthand engagements with accessibility challenges. Many described a mindset shift, underscoring that empathy-building is as critical as technical evaluation when advocating systemic change. A key limitation was the testing environment, while immersive, doesn’t fully mirror the unpredictability of public settings. Additionally, feedback from sighted users, though valuable, could not replicate the lived experience of blindness. The absence of post-exhibition interviews with PVI—after re-entering the room with vision—also limited the comparative insights between visual, tactile, and auditory cues.

Future work will focus on real-world trials. We are collaborating with NGOs to pilot the Tactosketchboard and Magic Platform in classrooms and public libraries in New Delhi, India. These long-term deployments will enable longitudinal insights and iterative refinement of the designs. We also aim to co-design with users across sensory and cognitive disabilities, addressing the underexplored intersections of access. As designers, researchers, and citizens, we must create with the body, through the senses, and for the dignity of all.

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